

Towfiqul Islam

Senior Software Engineer | Java / Spring Boot | Dhaka, Bangladesh

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Portfolio: <https://towfiq-ul.github.io>

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Professional Summary

Senior Software Engineer with 7+ years designing and shipping high-throughput distributed systems for fintech and enterprise clients. Core expertise in Java (11–21) / Spring Boot microservices, event-driven architecture with Kafka, and cloud-native infrastructure on AWS and Kubernetes. Currently building core payment platform components as Augmented Staff at bKash Limited, Bangladesh's #1 Unicorn MFS, handling millions of daily transactions. Passionate about clean architecture, system reliability, and shipping things that scale.

Java / Spring Boot Highlights

- Contributed to architecture design and implementation of microservices for bKash Limited's core payment platform, handling millions of daily transactions
- Built and maintain scalable Java 11–21 / Spring Boot microservices exposing RESTful APIs, using Virtual Threads and reactive programming (WebFlux)
- Delivered bKash-FinRec, a financial reconciliation system with corporate LDAP authentication, custom AOP-based access-control annotations, and audit-event logging shipped to a SIEM endpoint
- Built PIN Reset MW, a Spring Boot SOAP+REST middleware bridging the Huawei CPS mobile-money core platform, with JAXB codegen from WSDL/XSD
- Improved response time by 40% through database query optimization and Redis caching; achieved 85%+ unit and integration test coverage with JUnit and Mockito
- Extended ThingsBoard (open-source IoT platform, merged contributor — PR #6614) into a multi-tenant IoT middleware handling millions of data points for a Swiss enterprise customer
- Established coding standards, development workflows, and CI/CD pipelines using Jenkins and Docker, improving team productivity by 30%; mentors junior engineers on best practices and code reviews

Core Expertise

- **Backend:** Java (11–21), Spring Boot, Spring MVC, Spring Security, Spring Integration, Spring Cloud, Microservices, REST APIs, gRPC, Domain-Driven Design, Event-Driven Design
- **Cloud & DevOps:** AWS (EC2, S3, SNS, SQS), Docker, Kubernetes, Jenkins, Azure DevOps, CI/CD
- **Databases:** MySQL, Oracle, PostgreSQL, DynamoDB, Cassandra, Redis, JPA / Hibernate, Spring Data
- **Messaging:** Kafka, RabbitMQ, ActiveMQ, AWS SNS/SQS
- **Testing:** JUnit, TestNG, Mockito, Rest Assured, Postman
- **Security & Auth:** OAuth2, JWT, KeyCloak, LDAP

Work Experience

Senior Software Engineer – Exabyting

May 2024 – Current

Augmented resource at bKash Limited. Designing microservices, scalable REST APIs, Redis caching, and CI/CD pipelines for a core payment platform handling millions of daily transactions. Mentors junior engineers on best practices and technical decisions.

Software Engineer – SELISE

November 2021 – April 2024

Extended ThingsBoard into a multi-tenant IoT middleware platform for a Swiss enterprise customer, modeling and optimizing time-series telemetry across PostgreSQL, TimescaleDB, and Cassandra at a scale of millions of data points. Orchestrated the Swisscom integration and led direct client-facing discussions.

Software Engineer – Exabyting

December 2019 – November 2021

Developed government and private Java web applications and services. Worked as partner team with bKash Limited's Solution Engineering Team. Planned development cycles, prepared release notes, and conducted deployments in an Agile/Scrum process.

Major Java Projects

- **bKash App** – Consumer mobile financial app; Java 11–21 / Spring Boot (WebFlux) microservices, DynamoDB, Kafka, AWS, gRPC, Kubernetes
- **bKash-FinRec** – Financial reconciliation system (BanglaQR recon) with LDAP auth, custom AOP access control, SIEM audit logging
- **PIN Reset MW** – Spring Boot SOAP+REST middleware bridging Huawei CPS, JAXB codegen from WSDL/XSD
- **Tesenso MW** – Multi-tenant IoT platform for an EV/aerospace client
- **Urstamm** – Blockchain-based supply chain system
- **FDPS** – Financial document processing with Apache Camel & Kafka
- **MULTIBANK** – Automated multi-bank integration with SFTP and maker-checker approval flow
- **AMS** – Airtime Management System; automated CSV/XLS/PDF generation, SFTP operations, maker-checker approval flow, TPS-limiting service with interval-based retry

Key Impacts

- **bKash App**: built and maintained services processing 2.5M+ transactions/day
- **AMS**: reduced HTTP 429 rate-limit exceptions by solo-designing and building a self-contained, single-responsibility TPS-limiting service with interval-based retry
- **Spring Boot 2→3 migration**: led the migration across production services (javax→jakarta, AWS SDK v1→v2, DynamoDB layer rewrite) with zero disruption to live transaction volume
- **Tesenso MW**: extended to handle millions of IoT data points across a multi-tenant platform for an EV/aerospace client
- **Urstamm**: delivered a 20% performance improvement on the blockchain data pipeline, including a 0.5GB legacy data migration to blockchain
- **API latency**: reduced latency by replacing REST with GraphQL for client-facing data-fetching endpoints, eliminating over-fetching and cutting round trips for nested resources
- **CI/CD**: cut build time from 8m 29s to 4m 56s (~42% faster) by implementing multithreaded build execution in the pipeline
- **Developer tooling**: built a Bash/shell pipeline streaming live logs from an AWS EC2 instance to developers' local machines, cutting log-check time by ~15 minutes per check

Education

B.Sc. in Computer Science & Engineering – International Islamic University Chittagong (2014 - 2019)

Awards & Achievements

- IIUC IUPC 2017 – 2nd Position (Coach)
- EDU Programming Contest 2018 – 5th Position (Coach)
- CSE Week Programming Contest 2016 – 5th Position
- Solved 500+ competitive programming problems

Open Source Contributions

- ThingsBoard (Java, open-source IoT platform) – Pull Request #6614 (Merged)
- Tech Companies in Bangladesh – Pull Request #44 (Merged)
- jmbus (Java) – Author & Maintainer; JNA wrapper around the M-Bus (Meter-Bus) protocol library libmbus, grown out of Tesenso IoT platform work
- claude-auto-resume-command – Author & Maintainer; Claude Code /auto-resume slash command plus a Stop-hook-based session guardian that auto-resumes paused sessions